

Predictive Maintenance Agent for Manufacturing

System Architecture and Prototype Design

Architecture goal: Build a clear pipeline that takes machine/process data and turns it into a predictive maintenance decision.

1. High-Level System Architecture

Machine / Sensor Data → Data Collection → Data Processing → Condition Engine + ML Prediction Engine → Explainability / Failure Analysis → Maintenance Recommendation → Dashboard / Agent

2. Architecture Components

1. Data Collection

Receives machine/process readings such as temperature, rotational speed, torque and tool wear. In a prototype, this can be a CSV/API input. In a real factory, this layer can receive data from connected machines or an industrial data gateway.

2. Data Processing

Cleans and validates incoming values, handles missing values, selects required features and calculates derived parameters such as temperature difference, mechanical power and overstrain measure.

3. Condition Engine

Applies interpretable engineering conditions to identify abnormal operating situations and provide supporting evidence for the ML prediction.

4. ML Prediction Engine

Uses a trained classification model to calculate failure risk. A separate model or classification stage can identify the likely failure mode.

5. Explainability / Failure Analysis

Combines model output and condition evidence to explain why the machine is considered at risk. The output should use terms such as “likely failure mode” when exact physical causality cannot be proven.

6. Maintenance Recommendation Engine

Maps the predicted condition/failure mode to a practical maintenance action and urgency level.

7. Dashboard / Agent

Shows machine health, risk score, likely failure mode, supporting parameters, recommendation and alert status in a simple interface.

3. Processing Flow

- Step 1 — Receive machine readings.
- Step 2 — Validate and preprocess the data.
- Step 3 — Calculate derived parameters.

- Step 4 — Run condition checks.
- Step 5 — Send features to the trained ML model.
- Step 6 — Generate failure probability and predicted failure mode.
- Step 7 — Combine prediction with supporting evidence.
- Step 8 — Generate maintenance recommendation.
- Step 9 — Display the result and raise an alert when required.

4. Example Decision Flow

Input: Air temperature, process temperature, rotational speed, torque, tool wear and product type.

Processing: Calculate temperature difference, mechanical power and overstrain measure.

Prediction: ML model returns a high failure probability.

Analysis: High torque and high tool wear support an overstrain warning.

Agent output: “High failure risk. Likely failure mode: Overstrain Failure. Evidence: high torque and high tool wear. Recommended action: inspect tool condition and machine load.”

5. Prototype Technology Structure

Layer	Prototype implementation
Data	CSV dataset / simulated incoming readings
Processing	Python data-processing pipeline
ML	Classification model trained on historical data
Condition Analysis	Python rule/condition layer
Recommendation	Rule-based mapping from risk/failure mode to action
Interface	Simple web dashboard or application screen
Alerts	Application notification or workflow trigger

6. ML Model Pipeline

- Load and inspect the historical data.
- Select the relevant input features.
- Prepare the target failure label.
- Split data into training and testing sets.
- Train a classification model.
- Evaluate precision, recall, F1-score and confusion matrix.
- Save the trained model.
- Use the saved model for new machine readings.

7. Failure Analysis Layer

The analysis layer should not depend only on the ML prediction. It should also calculate interpretable indicators. This creates a hybrid design in which the ML model answers “how likely is failure?” while the condition layer helps answer “what operating condition supports this result?”

8. Dashboard Layout

Section	Information shown
Machine Status	Healthy / Warning / High Risk
Risk	Failure probability or risk score
Failure Mode	Most likely failure mode
Evidence	Important sensor/derived values
Recommendation	Suggested maintenance action
History	Recent risk/prediction trend when historical timestamps are available
Alert	Maintenance attention required

9. Service / Workflow Integration Concept

If the prototype is implemented on an enterprise workflow platform, the prediction can create a maintenance alert or work item. The workflow can assign the issue to the responsible maintenance team, record the recommendation, track acknowledgement and maintain a history of actions. The AI prediction remains the analytical layer; workflow automation handles the operational response.

10. Future Real-Factory Architecture

Machines / Sensors → Industrial Data Gateway → Data Platform → Feature Processing → ML + Condition Engine → Prediction Service → Maintenance Workflow → Technician → Feedback / Maintenance History → Model Improvement

This extended architecture requires real-time machine connectivity, timestamps, machine identifiers, maintenance records and feedback from completed maintenance activities.

11. Final Prototype Architecture

For the hackathon prototype, keep the implementation focused:

- Historical machine data as the input.
- Feature/derived-parameter processing.
- ML failure prediction.
- Failure-mode analysis.
- Maintenance recommendation.
- Simple dashboard/agent output.

This is enough to demonstrate the complete predictive-maintenance workflow without unnecessarily expanding the prototype into a full industrial IoT platform.